MODULE 2 — Conceptual Modeling for Programmers

In computational science, a **conceptual model** is a structured representation of a real-world system that highlights its **essential components, interactions, and rules** while ignoring irrelevant details. Conceptual modeling is the first and critical step before any coding begins because it defines the **scope and logic of the simulation**. Without a clear conceptual model, programmers may waste time coding unnecessary features, misrepresent the system, or produce a simulation that does not provide meaningful insights.

From a **computer science perspective**, conceptual modeling is analogous to **designing data structures, classes, and algorithms** before implementation. Entities in the system are modeled as **objects or agents**, variables describe the **state of each object**, and rules govern **how entities interact or evolve over time**. For instance, in a traffic flow simulation:

- **Entities**: Cars, traffic lights, pedestrians
- **Variables (states)**: Speed, position, direction, traffic light status
- **Rules (behaviors)**: Cars accelerate or decelerate based on traffic, stop at red lights, or maintain distance from other cars

By clearly defining these components, students learn how to **translate a complex, dynamic system into a computationally manageable framework**. Conceptual modeling also helps students to:

1. Identify **simplifying assumptions**, such as ignoring weather effects or considering uniform car size, so the simulation is feasible.
2. Recognize **interdependencies**, like how traffic light timing affects car movement.
3. Plan for **extensions or variations**, such as adding pedestrian interactions or adaptive traffic lights.

A well-thought-out conceptual model ensures that the simulation is **both accurate enough to produce meaningful patterns** and **simple enough to be implemented efficiently**. It also serves as a **communication tool**, allowing others (peers or instructors) to understand the student's system logic before code is written.

6. **Reflect on Scope and Feasibility**
    - Briefly describe **what is included** in the model and **what is excluded.**
    - Justify choices based on **importance, computational feasibility, and learning goals.**

Deliverables

1. **Conceptual Diagram**
    - A visual representation of entities, interactions, and key states.
    - Can be hand-drawn or created using diagramming software.
2. **Written Justification (1–2 paragraphs)**
    - Explain the entities, variables, rules, and assumptions.
    - Justify why the system is modeled this way and how simplifications help focus the simulation.
3. This rubric evaluates students' ability to **analyze a real-world system, identify entities, variables, and rules, and produce a clear conceptual model** that can serve as the foundation for a computational simulation.

Criteria	Beginning (1)	Developing (2)	Proficient (3)	Strong (4)
Identification of Entities	Few or unclear entities; major components of the system are missing	Some key entities identified, but some are unclear or incomplete	Most key entities identified clearly	All major entities identified, clearly defined, and relevant
Identification of Variables / States	Variables are missing, irrelevant, or poorly described	Some key variables described, but some unclear or incomplete	Most relevant variables are described and linked to entities	All relevant variables clearly described, linked to entities, and logically complete

Activity — Module 2: Conceptual Model Creation

Objective: Build a detailed conceptual model of your chosen problem, including entities, variables, interactions, and assumptions. This forms the **blueprint** for your simulation.

Step-by-Step Instructions for Students:

1. **Identify Entities**
   - List all key components that make up the system.
   - Example (traffic flow): Cars, pedestrians, traffic lights, intersections.
2. **List Variables (States) for Each Entity**
   - Identify the properties or attributes that describe the system at any given time.
   - Example (cars): Position, speed, direction, acceleration.
   - Example (traffic lights): Status (red, yellow, green), timer.
3. **Define Rules and Behaviors**
   - Specify how entities interact with each other and respond to environmental conditions.
   - Example (cars): Accelerate if road ahead is clear; brake if another car is too close; stop at red lights.
   - Example (traffic lights): Cycle through red → green → yellow; affect car behavior accordingly.
4. **Draw a Conceptual Diagram**
   - Use boxes or circles for entities and arrows for interactions or dependencies.
   - Highlight **key states and rules** visually for clarity.
5. **Document Assumptions and Simplifications**
   - List all **decisions that simplify the real-world system.**
   - Examples: Ignore weather, assume constant car size, no road accidents, uniform pedestrian behavior.
   - Explain why these assumptions are reasonable and how they help make simulation feasible.

Definition of Rules / Behaviors	Rules missing, vague, or inconsistent with the system	Some rules defined, but incomplete or partially inconsistent	Most rules clearly defined and consistent with system logic	All rules fully defined, logically consistent, and reflect meaningful system behavior
Conceptual Diagram / Visualization	Diagram missing, unclear, or disorganized	Diagram partially represents entities and interactions; some clarity issues	Diagram clearly represents entities, variables, and interactions	Diagram is clear, well-organized, annotated, and fully represents entities, variables, and interactions
Assumptions and Simplifications	Assumptions unclear, missing, or unrealistic	Some assumptions stated, but limited explanation or relevance	Most assumptions clearly stated and justified	All assumptions clearly stated, justified, and demonstrate understanding of model scope and simplifications
Clarity and Coherence of Explanation	Explanation poorly written, disorganized, or hard to follow	Explanation somewhat clear; some logic missing	Explanation clear and logically organized	Explanation is concise, well-written, fully logical, and communicates reasoning effectively